

EFY Note

The source code of this project is available for free download at source.efymag.com

```
## dronevoicecontrol.py
from aiy.board import Board, Led
import speech_recognition as sr
from tello_python import tello

import math
import time
import time
r = sr.Recognizer()
drone = tello.Tello()

def main():
    print("LED is ON while button is pressed
          (Ctrl-C for exit).")
    with Board() as board:
        while True:
            board.button.wait_for_press()
            with sr.Microphone() as source:
                r.adjust_for_ambient_noise(source)
                print("Say something!")
                audio = r.listen(source)
                command=str(r.recognize_
                           google(audio))
                print("Google Speech Recognition
                      thinks you said " + command)
            if (command=="Take off"):
                print("taking off drone")
                drone.takeoff()

            if (command=="move forward"):
                print("landing drone")
                drone.forward(100)

            if (command=="move backward"):
                print("landing drone")
                drone.backward(30)

            if (command== "Move right"):
                print("Moving drone right")
                drone.right(70)

            if (command== "move left"):
                print("Moving drone left")
                drone.forward(70)

            if (command== "move up"):
                print("moving drone up")
                drone.forward(100)

if __name__ == '__main__':
    main()
```

Fig. 3: Python code

associated libraries in the Python code (*dronevoicecontrol.py*). To con-



Fig. 4: Raspberry Pi connected to laptop through Ethernet cable

trol the drone, import the TelloPython library in the code shown in Fig. 3.

After doing all of the above, create a while loop, which will run continuously until stopped. This while loop assists in checking that when the bonnet button is pressed, the voice command gets processed.

To execute an action according to the command given by the user, create several if conditions. So, suppose the user wishes the drone to start flying. By giving voice command, say "take off," the request is matched with the if condition. If it matches the action set in the code, a signal is sent to the Tello drone to start flying.

Similarly, you can set all other functions in the code given here to command the drone to do the corresponding actions, such as move up, down, right, left, forward, etc.

Testing

Connect the AIY Voice Bonnet to Raspberry Pi board. Then connect

Raspberry Pi to an HDMI display and the Internet through an Ethernet cable or USB tethering. (The author's Raspberry Pi board connected to laptop through Ethernet cable is shown in Fig. 4.) Switch on the drone control device (Raspberry Pi) and then connect it to the Tello Wi-Fi. Now you can control the Tello drone through Wi-Fi network.

When all the connections have been made, run the Python code. A video of the drone surroundings is displayed on the Raspberry Pi's monitor screen. Now you may press the Bonnet button on AIY Voice Bonnet board and give voice commands to the drone and make it fly without any remote.

Congratulations! You have successfully made a drone that can be controlled by voice commands. **EFY**

Ashwini Kumar Sinha is an electronics hobbyist and tech journalist at EFY

Visit 'Online Only' expo-cum-conference from anywhere, any time.

 **TechWorldCongress**
@Home 2021

MONTHLY EVENT. FREE ENTRY. REGISTER NOW.

 | TechWorldCongress.com